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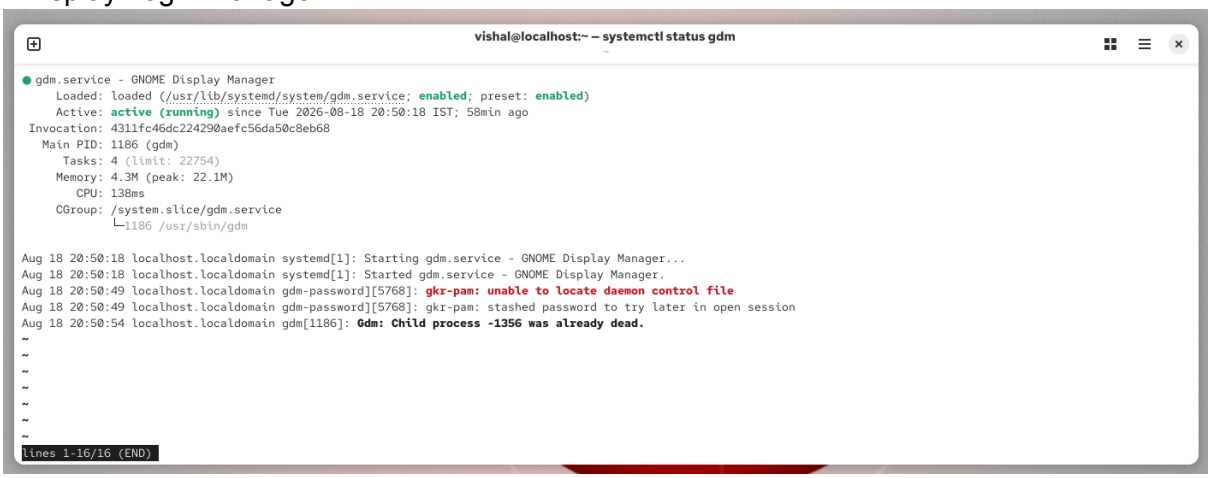
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Module 1

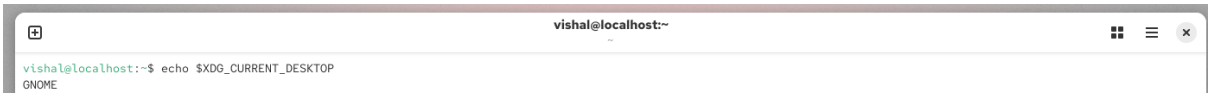
Practical 1:

a. Installing Red Hat Enterprise Linux (Virtual Box/ VmWare)
b. Linux Graphical Environment

i. Display/Login Manager



ii. Desktop Environment



iii. Linux Directory Structure and File Systems



Practical 2:

Aim: Basic System Commands

a. Use input-output redirection (>, >>, |, 2>, etc)

```
vishal@localhost:~$ echo "Hello Linux" > test.txt
vishal@localhost:~$ cat test.txt
Hello Linux
vishal@localhost:~$ echo "Welcome to RHEL 10" >> test.txt
vishal@localhost:~$ cat test.txt
Hello Linux
Welcome to RHEL 10
vishal@localhost:~$ ls | less
vishal@localhost:~$ ls | grep home
vishal@localhost:~$ ls /wrongfolder 2> error.txt
vishal@localhost:~$ cat error.txt
ls: cannot access '/wrongfolder': No such file or directory
```

b. Use grep and regular expressions to analyse text

```
vishal@localhost:~$ whoami
vishal
vishal@localhost:~$ sudo useradd student
useradd: user 'student' already exists
vishal@localhost:~$ [sudo] password for vishal:
bash: [sudo]: command not found...
vishal@localhost:~$ whoami
vishal
vishal@localhost:~$ sudo whoami
root
vishal@localhost:~$ sudo useradd student
useradd: user 'student' already exists
vishal@localhost:~$ sudo passwd student
New password:
Retype new password:
passwd: password updated successfully
vishal@localhost:~$ su - student
Password:
Last failed login: Tue Aug 18 22:44:39 IST 2026 on pts/0
There was 1 failed login attempt since the last successful login.
student@localhost:~$ whoami
student
student@localhost:~$ exit
logout
vishal@localhost:~$ whoami
vishal
```

```
vishal@localhost:~$ cat > students.txt
Vishal BScIT
Rahul BScIT
Priya BCom
Amit BScIT
Sneha BCA
Rohan BScITvishal@logrep "BScIT" students.txtudents.txt
Vishal BScIT
Rahul BScIT
Amit BScIT
Rohan BScIT
vishal@localhost:~$ grep "^V" students.txt
Vishal BScIT
vishal@localhost:~$ grep "IT$" students.txt
Vishal BScIT
Rahul BScIT
Amit BScIT
Rohan BScIT
```

c. Log in and switch users in multi-user targets

```
vishal@localhost:~$ whoami
vishal
vishal@localhost:~$ sudo useradd student
useradd: user 'student' already exists
vishal@localhost:~$ [sudo] password for vishal:
bash: [sudo]: command not found...
vishal@localhost:~$ whoami
vishal
vishal@localhost:~$ sudo whoami
root
vishal@localhost:~$ sudo useradd student
useradd: user 'student' already exists
vishal@localhost:~$ sudo passwd student
New password:
Retype new password:
passwd: password updated successfully
vishal@localhost:~$ su - student
Password:
Last failed login: Tue Aug 18 22:44:39 IST 2026 on pts/0
There was 1 failed login attempt since the last successful login.
student@localhost:~$ whoami
student
student@localhost:~$ exit
logout
vishal@localhost:~$ whoami
vishal
```

d. Working with Text Files cat,cut,less,hesd,tail,sort,wc,grep, awk

```
vishal@localhost:~$ cat students.txt
Vishal BScIT
Rahul BScIT
Priya BCom
Amit BScIT
Sneha BCA
Rohan BScITvishal@localhost:~$ head students.txt
Vishal BScIT
Rahul BScIT
Priya BCom
Amit BScIT
Sneha BCA
Rohan BScITvishal@localhost:~$ tail students.txt
Vishal BScIT
Rahul BScIT
Priya BCom
Amit BScIT
Sneha BCA
Rohan BScITvishal@localhost:~$ sort students.txt
Amit BScIT
Priya BCom
Rahul BScIT
Rohan BScIT
Sneha BCA
Vishal BScIT
vishal@localhost:~$ wc students.txt
 5 12 68 students.txt
vishal@localhost:~$ wc -l students.txt
5 students.txt
vishal@localhost:~$ grep "BScIT" students.txt
Vishal BScIT
Rahul BScIT
Amit BScIT
Rohan BScIT
vishal@localhost:~$ cat > data.txt
Vishal,BScIT,Mumbai
Rahul,BScIT,Pune
Priya,BCom,Mumbai
Amit,BScIT,Thane
Sneha,BCA,Nashik
vishal@localhost:~$ cut -d',' -f1 data.txt
Vishal
Rahul
Priya
Amit
Sneha
vishal@localhost:~$ less data.txt
vishal@localhost:~$ awk -F',' '{print $1}' data.txt
Vishal
Rahul
Priya
Amit
```

Practical 3:

Aim: Essential File Management Tools

a. Managing Files ,Using Links

```
vishal@localhost:~/practical3$ echo "This is my original file" > original.txt
vishal@localhost:~/practical3$ ln original.txt hardlink.txt
vishal@localhost:~/practical3$ ls -li
total 8
 68529742 -rw-r--r--. 1 vishal vishal  0 Aug 18 23:47 file1.txt
101376331 drwxr-xr-x. 2 vishal vishal 23 Aug 18 23:49 folder1
 68529745 -rw-r--r--. 2 vishal vishal 25 Aug 18 23:54 hardlink.txt
 68529745 -rw-r--r--. 2 vishal vishal 25 Aug 18 23:54 original.txt
vishal@localhost:~/practical3$ ln -s original.txt softlink.txt
vishal@localhost:~/practical3$ ls -l
total 8
-rw-r--r--. 1 vishal vishal  0 Aug 18 23:47 file1.txt
drwxr-xr-x. 2 vishal vishal 23 Aug 18 23:49 folder1
-rw-r--r--. 2 vishal vishal 25 Aug 18 23:54 hardlink.txt
-rw-r--r--. 2 vishal vishal 25 Aug 18 23:54 original.txt
lrwxrwxrwx. 1 vishal vishal 12 Aug 18 23:55 softlink.txt -> original.txt
```

b. Working with Archives and Compressed Files

```
vishal@localhost:~/practical3$ cd ~/practical3
vishal@localhost:~/practical3$ tar -cf sample.tar fileA.txt fileB.txt
vishal@localhost:~/practical3$ tar -tf sample.tar
fileA.txt
fileB.txt
vishal@localhost:~/practical3$ mkdir extracted
tar -xf sample.tar -C extracted
mkdir: cannot create directory ‘extracted’: File exists
vishal@localhost:~/practical3$ ls extracted
fileA.txt  fileB.txt
```

c. Create, delete, copy, and move files and directories

```
vishal@localhost:~$ mkdir ~/practical3
cd ~/practical3
vishal@localhost:~/practical3$ pwd
/home/vishal/practical3
vishal@localhost:~/practical3$ touch file1.txt
vishal@localhost:~/practical3$ ls
file1.txt
vishal@localhost:~/practical3$ mkdir folder1
vishal@localhost:~/practical3$ ls
file1.txt  folder1
vishal@localhost:~/practical3$ ls folder1
vishal@localhost:~/practical3$ touch file2.txt
vishal@localhost:~/practical3$ mv file2.txt folder1/
vishal@localhost:~/practical3$ ls folder1
file2.txt
vishal@localhost:~/practical3$ ls folder1
file2.txt
vishal@localhost:~/practical3$ cd ~/practical3
vishal@localhost:~/practical3$ touch delete-me.txt
vishal@localhost:~/practical3$ rm delete-me.txt
vishal@localhost:~/practical3$ ls
file1.txt  folder1
vishal@localhost:~/practical3$
```

d. Archive, compress, unpack, and uncompress files using tar, star, gzip, and bzip2

```
vishal@localhost:~/practical3$ tar -cvf files.tar fileA.txt fileB.txt
fileA.txt
fileB.txt
vishal@localhost:~/practical3$ tar -tvf files.tar
-rw-r--r-- vishal/vishal    32 2026-08-19 00:00 fileA.txt
-rw-r--r-- vishal/vishal    18 2026-08-19 00:00 fileB.txt
vishal@localhost:~/practical3$ mkdir tar_extract
tar -xvf files.tar -C tar_extract
fileA.txt
fileB.txt
vishal@localhost:~/practical3$ ls tar_extract
fileA.txt  fileB.txt
```

```
vishal@localhost:~/practical3$ gzip -k fileA.txt
vishal@localhost:~/practical3$ ls -lh fileA.txt.gz
-rw-r--r--. 1 vishal vishal 62 Aug 19 00:00 fileA.txt.gz
vishal@localhost:~/practical3$ gunzip fileA.txt.gz
gzip: fileA.txt already exists; do you wish to overwrite (y or n)? y
vishal@localhost:~/practical3$ bzip2 -k fileB.txt
vishal@localhost:~/practical3$ ls -lh fileB.txt.bz2
-rw-r--r--. 1 vishal vishal 63 Aug 19 00:00 fileB.txt.bz2
vishal@localhost:~/practical3$ bunzip2 fileB.txt.bz2
bunzip2: Output file fileB.txt already exists.
```

e. Create hard and soft links

```
vishal@localhost:~/practical3$ ln original.txt hardlink.txt
ln: failed to create hard link 'hardlink.txt': File exists
vishal@localhost:~/practical3$ ls
extracted  file1.txt  fileA.txt  fileB.txt  fileB.txt.bz2  files.tar  folder1  hardlink.txt  original.txt  sample.tar  softlink.txt  tar_extract
vishal@localhost:~/practical3$ ln -s original.txt softlink.txt
ln: failed to create symbolic link 'softlink.txt': File exists
vishal@localhost:~/practical3$ ls -l
total 44
drwxr-xr-x. 2 vishal vishal    40 Aug 19 00:17 extracted
-rw-r--r--. 1 vishal vishal     0 Aug 18 23:47 file1.txt
-rw-r--r--. 1 vishal vishal    32 Aug 19 00:00 fileA.txt
-rw-r--r--. 1 vishal vishal    18 Aug 19 00:00 fileB.txt
-rw-r--r--. 1 vishal vishal    63 Aug 19 00:00 fileB.txt.bz2
-rw-r--r--. 1 vishal vishal 10240 Aug 19 00:21 files.tar
drwxr-xr-x. 2 vishal vishal    23 Aug 18 23:49 folder1
-rw-r--r--. 2 vishal vishal    25 Aug 18 23:54 hardlink.txt
-rw-r--r--. 2 vishal vishal    25 Aug 18 23:54 original.txt
-rw-r--r--. 1 vishal vishal 10240 Aug 19 00:16 sample.tar
lrwxrwxrwx. 1 vishal vishal    12 Aug 18 23:55 softlink.txt -> original.txt
drwxr-xr-x. 2 vishal vishal    40 Aug 19 00:22 tar_extract
```

f. Managing File Ownership, Managing Basic Permissions

```
vishal@localhost:~/practical3$ ls -l original.txt
-rw-r--r--. 2 vishal vishal 25 Aug 18 23:54 original.txt
vishal@localhost:~/practical3$ sudo chown username original.txt
[sudo] password for vishal:
chown: invalid user: ‘username’
vishal@localhost:~/practical3$ chmod 644 original.txt
vishal@localhost:~/practical3$ ls -l original.txt
-rw-r--r--. 2 vishal vishal 25 Aug 18 23:54 original.txt
```

g. Managing Advanced Permissions, Managing ACLs

```
vishal@localhost:~$ cd ~/practical3
vishal@localhost:~/practical3$ ls
extracted  file1.txt  fileA.txt  fileB.txt  fileB.txt.bz2  files.tar  folder1  hardlink.txt  original.txt  practical4  sample.tar  softlink.txt  tar_extract
vishal@localhost:~/practical3$ touch aclfile.txt
vishal@localhost:~/practical3$ ls -l aclfile.txt
-rw-r--r--. 1 vishal vishal 0 Aug 19 01:26 aclfile.txt
vishal@localhost:~/practical3$ sudo setfacl -m g:students:rw aclfile.txt
vishal@localhost:~/practical3$ getfacl aclfile.txt
# file: aclfile.txt
# owner: vishal
# group: vishal
user::rw-
group::r--
group:students:rw-
mask::rw-
other::r--
```

h. Setting Default Permissions with umask

```
vishal@localhost:~/practical3$ umask
0022
vishal@localhost:~/practical3$ umask 022
vishal@localhost:~/practical3$
vishal@localhost:~/practical3$ umask
0022
```

i. Working with User-Extended Attributes

```
vishal@localhost:~/practical3$ touch attrfile.txt
vishal@localhost:~/practical3$ setfattr -n user.description -v "Practical3" attrfile.txt
vishal@localhost:~/practical3$ getfattr attrfile.txt
# file: attrfile.txt
user.description
```

j. List, set, and change standard ugo/rwx permissions

```
vishal@localhost:~/practical3$ ls -l attrfile.txt
-rw-r--r--. 1 vishal vishal 0 Aug 19 01:32 attrfile.txt
vishal@localhost:~/practical3$ chmod u=rwx,g=rx,o=r attrfile.txt
vishal@localhost:~/practical3$
vishal@localhost:~/practical3$ ls -l attrfile.txt
-rwxr-xr--. 1 vishal vishal 0 Aug 19 01:32 attrfile.txt
```

k. Create and configure set-GID directories for collaboration

```
vishal@localhost:~/practical3$ mkdir collaboration
vishal@localhost:~/practical3$ sudo chgrp students collaboration
[sudo] password for vishal:
vishal@localhost:~/practical3$ chmod g+s collaboration
vishal@localhost:~/practical3$ ls -ld collaboration
drwxr-xr-x. 2 vishal students 6 Aug 19 01:38 collaboration
```

l. Create and manage access control lists

```
vishal@localhost:~/practical3$ getfacl aclfile.txt
# file: aclfile.txt
# owner: vishal
# group: vishal
user::rw-
group::r--
group:students:rw-
mask::rw-
other::r--

vishal@localhost:~/practical3$ sudo setfacl -m g:students:rw aclfile.txt
vishal@localhost:~/practical3$ getfacl aclfile.txt
# file: aclfile.txt
# owner: vishal
# group: vishal
user::rw-
group::r--
group:students:rw-
mask::rw-
other::r--
```

Practical 4:

Aim: Working with Users, Groups, and Permissions

a. Creating and Managing Users

```
vishal@localhost:~/practical3$ sudo adduser testuser
[sudo] password for vishal:
vishal@localhost:~/practical3$ id testuser
uid=1002(testuser) gid=1002(testuser) groups=1002(testuser)
vishal@localhost:~/practical3$ sudo passwd testuser
New password:
Retype new password:
passwd: password updated successfully
```

b. Creating and Managing Groups

```
vishal@localhost:~/practical4$ sudo groupadd students
[sudo] password for vishal:
vishal@localhost:~/practical4$ sudo usermod -aG students vishal
vishal@localhost:~/practical4$ groups vishal
vishal : vishal wheel students
```

c. Using Permissions and Advanced Permissions

```
vishal@localhost:~/practical4$ touch testfile.txt
vishal@localhost:~/practical4$ ls -l testfile.txt
-rw-r--r--. 1 vishal vishal 0 Aug 19 00:54 testfile.txt
vishal@localhost:~/practical4$ chmod 644 testfile.txt
vishal@localhost:~/practical4$ ls -l testfile.txt
-rw-r--r--. 1 vishal vishal 0 Aug 19 00:54 testfile.txt
vishal@localhost:~/practical4$ setfacl --help
setfacl 2.3.2 -- set file access control lists
Usage: setfacl [-bkndRLP] { -m|-M|-x|-X ... } file ...
  -m, --modify=acl        modify the current ACL(s) of file(s)
  -M, --modify-file=file  read ACL entries to modify from file
  -x, --remove=acl        remove entries from the ACL(s) of file(s)
  -X, --remove-file=file  read ACL entries to remove from file
  -b, --remove-all        remove all extended ACL entries
  -k, --remove-default    remove the default ACL
      --set=acl           set the ACL of file(s), replacing the current ACL
      --set-file=file     read ACL entries to set from file
      --mask              do recalculate the effective rights mask
  -n, --no-mask           don't recalculate the effective rights mask
  -d, --default           operations apply to the default ACL
  -R, --recursive         recurse into subdirectories
  -L, --logical           logical walk, follow symbolic links
  -P, --physical          physical walk, do not follow symbolic links
      --restore=file      restore ACLs (inverse of `getfacl -R')
      --test              test mode (ACLs are not modified)
  -v, --version           print version and exit
  -h, --help              this help text
vishal@localhost:~/practical4$ sudo setfacl -m g:students:rw testfile.txt
[sudo] password for vishal:
vishal@localhost:~/practical4$ getfacl testfile.txt
# file: testfile.txt
# owner: vishal
# group: vishal
user::rw-
group::r--
group:students:rw-
mask::rw-
other::r--
```

d. Change passwords and adjust password aging for local user Accounts

```
vishal@localhost:~/practical4$ sudo passwd vishal
New password:
Retype new password:
passwd: password updated successfully
vishal@localhost:~/practical4$ sudo chage -M 90 vishal
vishal@localhost:~/practical4$ sudo chage -m 1 vishal
vishal@localhost:~/practical4$ sudo chage -W 7 vishal
vishal@localhost:~/practical4$ sudo chage -l vishal
Last password change                                    : Aug 18, 2026
Password expires                                        : Nov 16, 2026
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 1
Maximum number of days between password change          : 90
Number of days of warning before password expires       : 7
```

e. Configure superuser access -> su,sudo,root

```
vishal@localhost:~$ sudo passwd root
[sudo] password for vishal:
New password:
Retype new password:
passwd: password updated successfully
vishal@localhost:~$ su -
Password:
Last failed login: Wed Aug 19 01:07:01 IST 2026 on pts/0
There were 3 failed login attempts since the last successful login.
root@localhost:~# whoami
root
```

Practical 5

Aim: Managing Software

a. Managing Software Packages with YUM

```
vishal@localhost:~/practical6$ cd ~
vishal@localhost:~$ mkdir practical5
vishal@localhost:~$ cd practical5
vishal@localhost:~/practical5$ yum --version
4.20.0
  Installed: dnf-0:4.20.0-12.el10_0.noarch at Tuesday 18 August 2026 03:16:19 PM
  Built    : Red Hat, Inc. <http://bugzilla.redhat.com/bugzilla> at Monday 10 March 2025 10:02:24 PM

  Installed: rpm-0:4.19.1.1-12.el10.x86_64 at Tuesday 18 August 2026 03:14:54 PM
  Built    : Red Hat, Inc. <http://bugzilla.redhat.com/bugzilla> at Friday 07 February 2025 02:59:06 PM
vishal@localhost:~/practical5$ yum list installed
Not root, Subscription Management repositories not updated
Installed Packages
ModemManager.x86_64                                  1.22.0-7.el10                          @anaconda
ModemManager-glib.x86_64                             1.22.0-7.el10                          @anaconda
NetworkManager.x86_64                                1:1.52.0-1.el10_0                      @anaconda
NetworkManager-adsl.x86_64                           1:1.52.0-1.el10_0                      @anaconda
NetworkManager-bluetooth.x86_64                      1:1.52.0-1.el10_0                      @anaconda
NetworkManager-config-server.noarch                  1:1.52.0-1.el10_0                      @anaconda
NetworkManager-libnm.x86_64                          1:1.52.0-1.el10_0                      @anaconda
NetworkManager-tui.x86_64                            1:1.52.0-1.el10_0                      @anaconda
NetworkManager-wifi.x86_64                           1:1.52.0-1.el10_0                      @anaconda
NetworkManager-wwan.x86_64                           1:1.52.0-1.el10_0                      @anaconda
PackageKit.x86_64                                    1.2.8-8.el10                           @AppStream
PackageKit-command-not-found.x86_64                  1.2.8-8.el10                           @AppStream
PackageKit-glib.x86_64                               1.2.8-8.el10                           @AppStream
PackageKit-gstreamer-plugin.x86_64                   1.2.8-8.el10                           @AppStream
PackageKit-gtk3-module.x86_64                        1.2.8-8.el10                           @AppStream
aardvark-dns.x86_64                                  2:1.14.0-1.el10                        @AppStream
accountsservice.x86_64                               23.13.9-6.el10                         @AppStream
accountsservice-libs.x86_64                          23.13.9-6.el10                         @AppStream
acl.x86_64                                           2.3.2-4.el10                           @anaconda
adcli.x86_64                                         0.9.2-9.el10                           @anaconda
adobe-mappings-cmap.noarch                           20230622-5.el10                        @AppStream
adobe-mappings-cmap-deprecated.noarch                20230622-5.el10                        @AppStream
adobe-mappings-pdf.noarch                            20190401-9.el10                        @AppStream
adwaita-cursor-theme.noarch                          46.0-3.el10                            @AppStream
adwaita-icon-theme.noarch                            46.0-3.el10                            @AppStream
alsa-lib.x86_64                                      1.2.13-2.el10                          @AppStream
alsa-ucm.noarch                                      1.2.13-2.el10                          @AppStream
alsa-utils.x86_64                                    1.2.13-1.el10                          @AppStream
alternatives.x86_64                                  1.30-2.el10                            @anaconda
amd-gpu-firmware.noarch                              20250314-15.el10                       @AppStream
amd-ucode-firmware.noarch                            20250314-15.el10                       @anaconda
appstream.x86_64                                     1.0.2-5.el10                           @AppStream
appstream-data.noarch                                1:10-20250213.1.el10                   @AppStream
at.x86_64                                            3.2.5-12.el10                          @anaconda
at-spi2-atk.x86_64                                   2.52.0-3.el10                          @AppStream
```

b. Using yum

```
Installed Packages
Name         : nano
Version      : 8.1
Release      : 3.el10
Architecture : x86_64
Size         : 2.9 M
Source       : nano-8.1-3.el10.src.rpm
Repository   : @System
From repo    : anaconda
Summary      : A small text editor
URL          : https://www.nano-editor.org
License      : GPL-3.0-or-later
Description  : GNU nano is a small and friendly text editor.
```

c. Managing Software Packages with RPM

```
vishal@localhost:~/practical5$ rpm -q nano
nano-8.1-3.el10.x86_64
vishal@localhost:~/practical5$ rpm -qi nano
Name        : nano
Version     : 8.1
Release     : 3.el10
Architecture: x86_64
Install Date: Tuesday 18 August 2026 08:46:55 PM
Group       : Unspecified
Size        : 2995356
License     : GPL-3.0-or-later
Signature   : RSA/SHA256, Wednesday 11 December 2024 08:00:35 PM, Key ID 199e2f91fd431d51
Source RPM  : nano-8.1-3.el10.src.rpm
Build Date  : Wednesday 11 December 2024 03:41:34 PM
Build Host  : x86-64-03.build.eng.rdu2.redhat.com
Packager    : Red Hat, Inc. <http://bugzilla.redhat.com/bugzilla>
Vendor      : Red Hat, Inc.
URL         : https://www.nano-editor.org
Summary     : A small text editor
Description :
GNU nano is a small and friendly text editor.
vishal@localhost:~/practical5$ rpm -ql nano
/etc/nanorc
/usr/bin/nano
/usr/bin/rnano
/usr/lib/.build-id
/usr/lib/.build-id/54
/usr/lib/.build-id/54/95db857d66a25253cdfdfd5d125ad20a334250
/usr/share/doc/nano
/usr/share/doc/nano/AUTHORS
/usr/share/doc/nano/ChangeLog
/usr/share/doc/nano/NEWS
/usr/share/doc/nano/README
/usr/share/doc/nano/THANKS
/usr/share/doc/nano/TODO
/usr/share/doc/nano/faq.html
/usr/share/doc/nano/nano.html
/usr/share/doc/nano/sample.nanorc
/usr/share/info/nano.info.gz
/usr/share/licenses/nano
/usr/share/licenses/nano/COPYING
/usr/share/locale/bg/LC_MESSAGES/nano.mo
/usr/share/locale/ca/LC_MESSAGES/nano.mo
/usr/share/locale/cs/LC_MESSAGES/nano.mo
/usr/share/locale/da/LC_MESSAGES/nano.mo
/usr/share/locale/de/LC_MESSAGES/nano.mo
/usr/share/locale/eo/LC_MESSAGES/nano.mo

/usr/share/nano/ada.nanorc
/usr/share/nano/asm.nanorc
/usr/share/nano/autoconf.nanorc
/usr/share/nano/awk.nanorc
/usr/share/nano/c.nanorc
/usr/share/nano/changelog.nanorc
/usr/share/nano/cmake.nanorc
/usr/share/nano/css.nanorc
/usr/share/nano/default.nanorc
/usr/share/nano/elisp.nanorc
/usr/share/nano/email.nanorc
/usr/share/nano/fortran.nanorc
/usr/share/nano/go.nanorc
/usr/share/nano/groff.nanorc
/usr/share/nano/guile.nanorc
/usr/share/nano/haskell.nanorc
/usr/share/nano/html.nanorc
/usr/share/nano/java.nanorc
/usr/share/nano/javascript.nanorc
/usr/share/nano/json.nanorc
/usr/share/nano/lua.nanorc
/usr/share/nano/makefile.nanorc
/usr/share/nano/man.nanorc
/usr/share/nano/markdown.nanorc
/usr/share/nano/nanohelp.nanorc
/usr/share/nano/nanorc.nanorc
/usr/share/nano/nftables.nanorc
/usr/share/nano/objc.nanorc
/usr/share/nano/ocaml.nanorc
/usr/share/nano/patch.nanorc
/usr/share/nano/perl.nanorc
/usr/share/nano/php.nanorc
/usr/share/nano/po.nanorc
/usr/share/nano/povray.nanorc
/usr/share/nano/python.nanorc
/usr/share/nano/ruby.nanorc
/usr/share/nano/rust.nanorc
/usr/share/nano/sh.nanorc
/usr/share/nano/spec.nanorc
/usr/share/nano/sql.nanorc
/usr/share/nano/tcl.nanorc
/usr/share/nano/tex.nanorc
/usr/share/nano/texinfo.nanorc
/usr/share/nano/xml.nanorc
/usr/share/nano/yaml.nanorc
vishal@localhost:~/practical5$ rpm -V nano
```

d. Install and update software packages repository from the local file system

```
vishal@localhost:~/practical5$ sudo nano /etc/yum.repos.d/local.repo
vishal@localhost:~/practical5$ sudo dnf repolist
[sudo] password for vishal:
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "subscription-manager" to register.

repo id                                                                                repo name
AppStream                                                                              RHEL 10 Local AppStream
BaseOS                                                                                 RHEL 10 Local BaseOS
vishal@localhost:~/practical5$ sudo dnf install nano
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "subscription-manager" to register.

RHEL 10 Local BaseOS                                                                   22 MB/s | 1.5 MB     00:00
RHEL 10 Local AppStream                                                                22 MB/s | 1.5 MB     00:00
Last metadata expiration check: 0:00:01 ago on Sunday 23 August 2026 01:24:35 AM.
Package nano-8.1-3.el10.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
vishal@localhost:~/practical5$ sudo dnf update
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "subscription-manager" to register.

Last metadata expiration check: 0:04:02 ago on Sunday 23 August 2026 01:24:35 AM.
Dependencies resolved.
Nothing to do.
Complete!
```

Module 2

Practical 6:

Aim: Managing Processes

a. Introduction to Process Management

b. Managing Shell Jobs

c. Using Common Command-Line Tools for Process Management

d. Using top to Manage Processes

e. Using tuned to Optimize Performance

f. Identify CPU/memory-intensive processes and kill processes

a. Introduction to Process Management

Aim: Introduction to process Management

A process is a running of a program. Every process has a unique **PID(Process ID)**

Useful concepts

Term	Meaning
PID	Unique ID assigned to process
PPID	Present Process ID
UID	User who owns the process
CPU%	Percentage of CPU being used
MEM%	Percentage of memory being used
Process State	Running,Sleeping,Stoped,Zombie,etc

View processes

1.Ps

```
vishal@localhost:~/practical6$ ps
    PID TTY          TIME CMD
   9801 pts/0    00:00:00 bash
  10555 pts/0    00:00:00 ps
```

2.Ps aux

```
vishal@localhost:~/practical6$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  1.1  49188 41380 ?        Ss   Aug18   0:06 /usr/lib/systemd/systemd --switched-root --system --deserialize=48 rhgb
root           2  0.0  0.0      0     0 ?        S    Aug18   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    Aug18   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   Aug18   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   Aug18   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   Aug18   0:00 [kworker/R-slub_flushwq]
root           7  0.0  0.0      0     0 ?        I<   Aug18   0:00 [kworker/R-netns]
root           9  0.0  0.0      0     0 ?        I<   Aug18   0:00 [kworker/0:0H-events_highpri]
root          11  0.0  0.0      0     0 ?        I    Aug18   0:00 [kworker/u512:0-events_unbound]
root          12  0.0  0.0      0     0 ?        I<   Aug18   0:00 [kworker/R-mm_percpu_wq]
root          13  0.0  0.0      0     0 ?        I    Aug18   0:00 [kworker/u512:1-ipv6_addrconf]
root          14  0.0  0.0      0     0 ?        I    Aug18   0:00 [rcu_tasks_kthread]
root          15  0.0  0.0      0     0 ?        I    Aug18   0:00 [rcu_tasks_rude_kthread]
root          16  0.0  0.0      0     0 ?        I    Aug18   0:00 [rcu_tasks_trace_kthread]
root          17  0.0  0.0      0     0 ?        S    Aug18   0:00 [ksoftirqd/0]
root          18  0.0  0.0      0     0 ?        I    Aug18   0:05 [rcu_preempt]
root          19  0.0  0.0      0     0 ?        S    Aug18   0:00 [rcu_exp_par_gp_kthread_worker/1]
root          20  0.0  0.0      0     0 ?        S    Aug18   0:00 [rcu_exp_gp_kthread_worker]
root          21  0.0  0.0      0     0 ?        S    Aug18   0:00 [migration/0]
root          22  0.0  0.0      0     0 ?        S    Aug18   0:00 [idle_inject/0]
root          23  0.0  0.0      0     0 ?        S    Aug18   0:00 [cpuhp/0]
root          24  0.0  0.0      0     0 ?        S    Aug18   0:00 [cpuhp/1]
root          25  0.0  0.0      0     0 ?        S    Aug18   0:00 [idle_inject/1]
root          26  0.0  0.0      0     0 ?        S    Aug18   0:00 [migration/1]
root          27  0.0  0.0      0     0 ?        S    Aug18   0:00 [ksoftirqd/1]
root          29  0.0  0.0      0     0 ?        I<   Aug18   0:00 [kworker/1:0H-events_highpri]
root          34  0.0  0.0      0     0 ?        S    Aug18   0:00 [kdevtmpfs]
root          35  0.0  0.0      0     0 ?        I<   Aug18   0:00 [kworker/R-inet_frag_wq]
root          37  0.0  0.0      0     0 ?        S    Aug18   0:00 [kauditd]
root          38  0.0  0.0      0     0 ?        S    Aug18   0:00 [khungtaskd]
root          40  0.0  0.0      0     0 ?        S    Aug18   0:00 [oom_reaper]
root          41  0.0  0.0      0     0 ?        I<   Aug18   0:00 [kworker/R-writeback]
root          42  0.0  0.0      0     0 ?        S    Aug18   0:00 [kcompactd0]
root          43  0.0  0.0      0     0 ?        SN   Aug18   0:00 [ksmd]
root          44  0.0  0.0      0     0 ?        SN   Aug18   0:04 [khugepaged]
root          45  0.0  0.0      0     0 ?        I<   Aug18   0:00 [kworker/R-cryptd]
```

3.pstree

```
vishal@localhost:~/practical6$ pstree
systemd─┬─ModemManager───3*[{ModemManager}]
        ├─NetworkManager───3*[{NetworkManager}]
        ├─VGAuthService
        ├─accounts-daemon───3*[{accounts-daemon}]
        ├─alsactl
        ├─atd
        ├─auditd─┬─sedispatch
        │        └─2*[{auditd}]
        ├─avahi-daemon───avahi-daemon
        ├─chronyd
        ├─colord───3*[{colord}]
        ├─crond
        ├─cupsd
        ├─dbus-broker-lau───dbus-broker
        ├─firewalld───{firewalld}
        ├─fwupd───5*[{fwupd}]
        ├─gdm─┬─gdm-session-wor─┬─gdm-wayland-ses─┬─gnome-session-b───4*[{gnome-session-b}]
        │     │                 │                 └─3*[{gdm-wayland-ses}]
        │     │                 └─3*[{gdm-session-wor}]
        │     └─3*[{gdm}]
        ├─gnome-keyring-d───4*[{gnome-keyring-d}]
        ├─gpg-agent─┬─scdaemon───{scdaemon}
        │           └─{gpg-agent}
        ├─irqbalance───{irqbalance}
        ├─lsmd
        ├─mcelog
        ├─polkitd───3*[{polkitd}]
        ├─rhsm-service───{rhsm-service}
        ├─rhsmcertd───{rhsmcertd}
        ├─rsyslogd───2*[{rsyslogd}]
        ├─rtkit-daemon───2*[{rtkit-daemon}]
        ├─sshd
        ├─sssd_kcm
        ├─switcheroo-cont───3*[{switcheroo-cont}]
        ├─systemd─┬─(sd-pam)
        │         ├─at-spi-bus-laun─┬─dbus-broker-lau───dbus-broker
        │         │                 └─4*[{at-spi-bus-laun}]
        │         ├─at-spi2-registr───3*[{at-spi2-registr}]
        │         ├─catatonit
        │         ├─dbus-broker-lau───dbus-broker
        │         ├─dconf-service───3*[{dconf-service}]
        │         ├─evolution-addre───6*[{evolution-addre}]
        │         ├─evolution-calen───9*[{evolution-calen}]
        │         ├─evolution-sourc───4*[{evolution-sourc}]
        │         ├─2*[gjs───11*[{gjs}]]
        │         ├─gnome-session-b─┬─evolution-alarm───8*[{evolution-alarm}]
        │         ├─gsd-power───5*[{gsd-power}]
        │         ├─gsd-print-notif───3*[{gsd-print-notif}]
        │         ├─gsd-printer───3*[{gsd-printer}]
        │         ├─gsd-rfkill───3*[{gsd-rfkill}]
        │         ├─gsd-screensaver───3*[{gsd-screensaver}]
        │         ├─gsd-sharing───4*[{gsd-sharing}]
        │         ├─gsd-smartcard───4*[{gsd-smartcard}]
        │         ├─gsd-sound───4*[{gsd-sound}]
        │         ├─gsd-subman───5*[{gsd-subman}]
        │         ├─gsd-usb-protect───4*[{gsd-usb-protect}]
        │         ├─gsd-wacom───5*[{gsd-wacom}]
        │         ├─gsd-xsettings───5*[{gsd-xsettings}]
        │         ├─gvfs-goa-volume───3*[{gvfs-goa-volume}]
        │         ├─gvfs-gphoto2-vo───3*[{gvfs-gphoto2-vo}]
        │         ├─gvfs-mtp-volume───3*[{gvfs-mtp-volume}]
        │         ├─gvfs-udisks2-vo───4*[{gvfs-udisks2-vo}]
        │         ├─gvfsd───3*[{gvfsd}]
        │         ├─gvfsd-fuse───6*[{gvfsd-fuse}]
        │         ├─gvfsd-metadata───3*[{gvfsd-metadata}]
        │         ├─ibus-daemon─┬─ibus-dconf───4*[{ibus-dconf}]
        │         │              ├─ibus-engine-sim───3*[{ibus-engine-sim}]
        │         │              ├─ibus-extension-───5*[{ibus-extension-}]
        │         │              └─3*[{ibus-daemon}]
        │         ├─ibus-portal───3*[{ibus-portal}]
        │         ├─ibus-x11───4*[{ibus-x11}]
        │         ├─pipewire───2*[{pipewire}]
        │         ├─pipewire-pulse───2*[{pipewire-pulse}]
        │         ├─ptyxis─┬─ptyxis-agent─┬─bash───pstree
        │         │        │              └─4*[{ptyxis-agent}]
        │         │        └─17*[{ptyxis}]
        │         ├─vmtoolsd───3*[{vmtoolsd}]
        │         ├─wireplumber───5*[{wireplumber}]
        │         ├─3*[xdg-desktop-por───5*[{xdg-desktop-por}]]
        │         ├─xdg-document-po─┬─fusermount3
        │         │                 └─7*[{xdg-document-po}]
        │         └─xdg-permission-───3*[{xdg-permission-}]
        ├─systemd-journal
        ├─systemd-logind
        ├─systemd-udevd
        ├─systemd-userdbd───3*[systemd-userwor]
        ├─tuned───3*[{tuned}]
        ├─tuned-ppd───3*[{tuned-ppd}]
        ├─udisksd───6*[{udisksd}]
        ├─upowerd───3*[{upowerd}]
        ├─vmtoolsd───3*[{vmtoolsd}]
        └─wpa_supplicant
```

b. Managing Shell Jobs

```
vishal@localhost:~/practical6$ sleep 60 &
[1] 10637
vishal@localhost:~/practical6$ jobs
[1]+  Running                 sleep 60 &
vishal@localhost:~/practical6$ fg
sleep 60
^Z
[1]+  Stopped                 sleep 60
vishal@localhost:~/practical6$ bg
[1]+ sleep 60 &
[1]+  Done                    sleep 60
```

c. Using Common Command-Line Tools for Process Management

```
vishal@localhost:~/practical6$ ps
    PID TTY          TIME CMD
   9801 pts/0    00:00:00 bash
  10663 pts/0    00:00:00 ps
vishal@localhost:~/practical6$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  1.1  49188 41380 ?        Ss   Aug18   0:06 /usr/lib/systemd/systemd --switched-root --system --deserialize=48 rhgb
root           2  0.0  0.0      0     0 ?        S    Aug18   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    Aug18   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   Aug18   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   Aug18   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   Aug18   0:00 [kworker/R-slub_flushwq]
root           7  0.0  0.0      0     0 ?        I<   Aug18   0:00 [kworker/R-netns]
root           9  0.0  0.0      0     0 ?        I<   Aug18   0:00 [kworker/0:0H-events_highpri]
root          11  0.0  0.0      0     0 ?        I    Aug18   0:00 [kworker/u512:0-events_unbound]
root          12  0.0  0.0      0     0 ?        I<   Aug18   0:00 [kworker/R-mm_percpu_wq]
root          13  0.0  0.0      0     0 ?        I    Aug18   0:00 [kworker/u512:1-ipv6_addrconf]
root          14  0.0  0.0      0     0 ?        I    Aug18   0:00 [rcu_tasks_kthread]
root          15  0.0  0.0      0     0 ?        I    Aug18   0:00 [rcu_tasks_rude_kthread]
root          16  0.0  0.0      0     0 ?        I    Aug18   0:00 [rcu_tasks_trace_kthread]
root          17  0.0  0.0      0     0 ?        S    Aug18   0:00 [ksoftirqd/0]
root          18  0.0  0.0      0     0 ?        R    Aug18   0:05 [rcu_preempt]
root          19  0.0  0.0      0     0 ?        S    Aug18   0:00 [rcu_exp_par_gp_kthread_worker/1]
root          20  0.0  0.0      0     0 ?        S    Aug18   0:00 [rcu_exp_gp_kthread_worker]
root          21  0.0  0.0      0     0 ?        S    Aug18   0:00 [migration/0]
root          22  0.0  0.0      0     0 ?        S    Aug18   0:00 [idle_inject/0]
```

```
vishal@localhost:~/practical6$ pgrep bash
9801
vishal@localhost:~/practical6$ pstree
systemd─┬─ModemManager───3*[{ModemManager}]
        ├─NetworkManager───3*[{NetworkManager}]
        ├─VGAuthService
        ├─accounts-daemon───3*[{accounts-daemon}]
        ├─alsactl
        ├─atd
        ├─auditd─┬─sedispatch
        │        └─2*[{auditd}]
        ├─avahi-daemon───avahi-daemon
        ├─chronyd
        ├─colord───3*[{colord}]
        ├─crond
        ├─cupsd
        ├─dbus-broker-lau───dbus-broker
        ├─firewalld───{firewalld}
        ├─fwupd───5*[{fwupd}]
```

```
vishal@localhost:~/practical6$ sleep 100 &
[1] 10682
vishal@localhost:~/practical6$ pgrep sleep
10682
vishal@localhost:~/practical6$ kill PID
bash: kill: PID: arguments must be process or job IDs
```

d. Using top to Manage Processes

```
top - 02:16:46 up  4:33,  2 users,  load average: 0.23, 0.17, 0.12
Tasks: 319 total,   1 running, 318 sleeping,   0 stopped,   0 zombie
%Cpu(s):  6.1 us,  3.3 sy,  0.0 ni, 89.7 id,  0.0 wa,  0.5 hi,  0.3 si,  0.0 st
MiB Mem :   3619.4 total,   1295.7 free,   1771.6 used,    815.4 buff/cache
MiB Swap:   4004.0 total,   4004.0 free,      0.0 used.   1847.8 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
9759	vishal	20	0	1945028	395064	103908	S	7.3	10.7	1:45.34	ptyxis
5938	vishal	20	0	4256992	454768	127544	S	6.0	12.3	8:42.44	gnome-shell
10408	root	20	0	0	0	0	I	2.3	0.0	0:06.67	kworker/u514:0-events_unbound
10093	root	20	0	0	0	0	I	1.7	0.0	0:05.27	kworker/u513:1-events_unbound
1008	root	20	0	536352	8488	6824	S	0.3	0.2	1:00.64	vmtoolsd
6092	vishal	20	0	509788	39660	29800	S	0.3	1.1	1:00.12	vmtoolsd
10227	root	20	0	0	0	0	I	0.3	0.0	0:08.46	kworker/u513:2-events_unbound
10725	**vishal**	**20**	**0**	**231624**	**5280**	**3232**	**R**	**0.3**	**0.1**	**0:00.13**	**top**
1	root	20	0	49188	41380	10340	S	0.0	1.1	0:06.37	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.02	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
9	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-events_highpri
11	root	20	0	0	0	0	I	0.0	0.0	0:00.00	kworker/u512:0-events_unbound
12	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
13	root	20	0	0	0	0	I	0.0	0.0	0:00.04	kworker/u512:1-ipv6_addrconf
14	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
15	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
16	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kthread
17	root	20	0	0	0	0	S	0.0	0.0	0:00.13	ksoftirqd/0
18	root	20	0	0	0	0	I	0.0	0.0	0:05.79	rcu_preempt
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_par_gp_kthread_worker/1
20	root	20	0	0	0	0	S	0.0	0.0	0:00.01	rcu_exp_gp_kthread_worker
21	root	rt	0	0	0	0	S	0.0	0.0	0:00.09	migration/0
22	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
23	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
24	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
25	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/1
26	root	rt	0	0	0	0	S	0.0	0.0	0:00.80	migration/1
27	root	20	0	0	0	0	S	0.0	0.0	0:00.13	ksoftirqd/1
29	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/1:0H-events_highpri
34	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kdevtmpfs
35	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-inet_frag_wq
37	root	20	0	0	0	0	S	0.0	0.0	0:00.02	kauditd
38	root	20	0	0	0	0	S	0.0	0.0	0:00.07	khungtaskd
40	root	20	0	0	0	0	S	0.0	0.0	0:00.00	oom_reaper
41	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-writeback

e. Using tuned to Optimize Performance

```
vishal@localhost:~/practical6$ systemctl status tuned
● tuned.service - Dynamic System Tuning Daemon
     Loaded: loaded (/usr/lib/systemd/system/tuned.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-08-18 20:50:19 IST; 5h 28min ago
 Invocation: d57cfe71159643918d4dfed7468a6e46
       Docs: man:tuned(8)
             man:tuned.conf(5)
             man:tuned-adm(8)
   Main PID: 1170 (tuned)
      Tasks: 4 (limit: 22754)
     Memory: 18.3M (peak: 19.8M)
        CPU: 6.058s
     CGroup: /system.slice/tuned.service
             └─1170 /usr/bin/python3 -Es /usr/sbin/tuned -l -P

Aug 18 20:50:18 localhost.localdomain systemd[1]: Starting tuned.service - Dynamic System Tuning Daemon...
Aug 18 20:50:19 localhost.localdomain systemd[1]: Started tuned.service - Dynamic System Tuning Daemon.
vishal@localhost:~/practical6$ tuned-adm active
Current active profile: balanced
vishal@localhost:~/practical6$ tuned-adm list
Available profiles:
- accelerator-performance     - Throughput performance based tuning with disabled higher latency STOP states
- aws                         - Optimize for aws ec2 instances
- balanced                    - General non-specialized tuned profile
- balanced-battery            - Balanced profile biased towards power savings changes for battery
- desktop                     - Optimize for the desktop use-case
- hpc-compute                 - Optimize for HPC compute workloads
- intel-sst                   - Configure for Intel Speed Select Base Frequency
- latency-performance         - Optimize for deterministic performance at the cost of increased power consumption
- network-latency             - Optimize for deterministic performance at the cost of increased power consumption, focused on low latency network performance
- network-throughput          - Optimize for streaming network throughput, generally only necessary on older CPUs or 40G+ networks
- optimize-serial-console     - Optimize for serial console use.
- powersave                   - Optimize for low power consumption
- throughput-performance      - Broadly applicable tuning that provides excellent performance across a variety of common server workloads
- virtual-guest               - Optimize for running inside a virtual guest
- virtual-host                - Optimize for running KVM guests
Current active profile: balanced
vishal@localhost:~/practical6$ sudo tuned-adm profile throughput-performance
[sudo] password for vishal:
vishal@localhost:~/practical6$ tuned-adm active
Current active profile: throughput-performance
```

f. Identify CPU/memory-intensive processes and kill processes

```
vishal@localhost:~/practical6$ ps aux --sort=-%cpu | head
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
vishal      5938  3.2 12.2 4263896 454704 ?      Ssl  Aug18   8:57 /usr/bin/gnome-shell
vishal      9759  2.6 10.6 1946272 395660 ?      Ssl  01:08   2:00 /usr/bin/ptyxis --gapplication-service
vishal      6453  0.4  0.3 388020 12092 ?        S<sl Aug18   1:15 /usr/bin/pipewire
root        1008  0.3  0.2 536352  8488 ?        Ssl  Aug18   1:01 /usr/bin/vmtoolsd
vishal      6092  0.3  1.0 509788 39660 ?        Sl   Aug18   1:01 /usr/bin/vmtoolsd -n vmusr --uinputFd 3
root       10803  0.3  0.0      0     0 ?        I    02:20   0:00 [kworker/u514:3-events_unbound]
root       10408  0.3  0.0      0     0 ?        I    01:42   0:08 [kworker/u514:0-events_unbound]
root       10227  0.3  0.0      0     0 ?        I    01:27   0:10 [kworker/u513:2-events_unbound]
root       10740  0.1  0.0      0     0 ?        I    02:19   0:00 [kworker/u513:3-events_unbound]
vishal@localhost:~/practical6$ ps aux --sort=-%mem | head
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
vishal      5938  3.2 12.2 4263896 454704 ?      Rsl  Aug18   8:59 /usr/bin/gnome-shell
vishal      9759  2.7 10.6 1946272 395660 ?      Ssl  01:08   2:01 /usr/bin/ptyxis --gapplication-service
vishal      6089  0.0  4.6 1398924 170808 ?      Sl   Aug18   0:02 /usr/bin/gnome-software --gapplication-service
vishal      6621  0.0  2.7 1296128 102908 ?      Sl   Aug18   0:00 /usr/libexec/mutter-x11-frames
root        6519  0.0  1.6 389852 62092 ?        Ssl  Aug18   0:06 /usr/bin/python3 /usr/libexec/rhsm-service
vishal      6612  0.0  1.6 1118172 59788 ?       Ssl  Aug18   0:00 /usr/libexec/xdg-desktop-portal-gnome
vishal      6508  0.0  1.4 941476 54104 ?        Ssl  Aug18   0:00 /usr/libexec/gsd-xsettings
root        1068  0.0  1.2 359780 44788 ?        Ssl  Aug18   0:00 /usr/bin/python3 -sP /usr/sbin/firewalld --nofork --nopid
vishal      6348  0.0  1.1 188868 42448 ?        S    Aug18   0:00 /usr/bin/Xwayland :0 -rootless -noreset -accessx -core -auth /run/user/1000/.mutter-Xwaylandauth.RSR4T3 -listenfd 4 -listenfd 5 -displayfd
6 -initfd 7 -byteswappedclients -enable-ei-portal
vishal@localhost:~/practical6$ sleep 1000 &
[1] 10854
vishal@localhost:~/practical6$ pgrep sleep
10854
vishal@localhost:~/practical6$ kill 1234
bash: kill: (1234) - No such process
vishal@localhost:~/practical6$ kill 10854
[1]+  Terminated              sleep 1000
vishal@localhost:~/practical6$ pgrep sleep
```

Practical 7

Aim: Managing Storage

a. Understanding MBR and GPT Partitions

```
vishal@localhost:~$ sudo fdisk -l /dev/nvme0n2
[sudo] password for vishal:
Disk /dev/nvme0n2: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual NVMe Disk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
vishal@localhost:~$ sudo fdisk /dev/nvme0n2

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0x28579f53.

Command (m for help): g
Created a new GPT disklabel (GUID: C81BA5CF-E695-485A-B04C-3DD64B17318E).

Command (m for help): p
Disk /dev/nvme0n2: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual NVMe Disk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: C81BA5CF-E695-485A-B04C-3DD64B17318E

Command (m for help): n
Partition number (1-128, default 1):
First sector (2048-20971486, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-20971486, default 20969471): +2G

Created a new partition 1 of type 'Linux filesystem' and of size 2 GiB.

Command (m for help): p
Disk /dev/nvme0n2: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual NVMe Disk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: C81BA5CF-E695-485A-B04C-3DD64B17318E

Device         Start     End Sectors Size Type
/dev/nvme0n2p1  2048 4196351 4194304   2G Linux filesystem


Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

vishal@localhost:~$ lsblk
NAME          MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sr0            11:0    1  7.9G  0 rom  /run/media/vishal/RHEL-10-0-BaseOS-x86_64
nvme0n1       259:0    0   50G  0 disk
├─nvme0n1p1   259:2    0  600M  0 part /boot/efi
├─nvme0n1p2   259:3    0    1G  0 part /boot
└─nvme0n1p3   259:4    0 48.4G  0 part
  ├─rhel-root 253:0    0 44.5G  0 lvm  /
  └─rhel-swap 253:1    0  3.9G  0 lvm  [SWAP]
nvme0n2       259:1    0   10G  0 disk
└─nvme0n2p1   259:5    0    2G  0 part
```

b. Managing Partitions and File Systems

```
vishal@localhost:~$ sudo mkfs.ext4 /dev/nvme0n2p1
[sudo] password for vishal:
mke2fs 1.47.1 (20-May-2024)
Creating filesystem with 524288 4k blocks and 131072 inodes
Filesystem UUID: c669ffd7-dd89-42c6-9e43-b4d86d760913
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
```

c. Mounting File Systems

```
vishal@localhost:~$ sudo mkdir /mnt/ext4disk
[sudo] password for vishal:
vishal@localhost:~$ sudo mount /dev/nvme0n2p1 /mnt/ext4disk
vishal@localhost:~$ df -h
Filesystem             Size  Used Avail Use% Mounted on
/dev/mapper/rhel-root   45G  4.7G   40G  11% /
devtmpfs               4.0M     0  4.0M   0% /dev
tmpfs                  1.8G   84K  1.8G   1% /dev/shm
efivarfs               256K   76K  176K  31% /sys/firmware/efi/efivars
tmpfs                  724M  9.7M  715M   2% /run
tmpfs                  1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
/dev/nvme0n1p2         960M  326M  635M  34% /boot
/dev/nvme0n1p1         599M  8.4M  591M   2% /boot/efi
tmpfs                  362M  136K  362M   1% /run/user/1000
/dev/sr0               7.9G  7.9G     0 100% /run/media/vishal/RHEL-10-0-BaseOS-x86_64
/dev/nvme0n2p1         2.0G   24K  1.8G   1% /mnt/ext4disk
```

d. List, create, delete partitions on MBR and GPT disks

```
vishal@localhost:~$ sudo fdisk -l /dev/nvme0n2
Disk /dev/nvme0n2: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual NVMe Disk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: C81BA5CF-E695-485A-B04C-3DD64B17318E

Device          Start     End Sectors Size Type
/dev/nvme0n2p1   2048 4196351 4194304   2G Linux filesystem
```

```
vishal@localhost:~$ sudo fdisk /dev/nvme0n2

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.


Command (m for help): n
Partition number (2-128, default 2):
First sector (4196352-20971486, default 4196352):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (4196352-20971486, default 20969471): +2G

Created a new partition 2 of type 'Linux filesystem' and of size 2 GiB.

Command (m for help): p
Disk /dev/nvme0n2: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual NVMe Disk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: C81BA5CF-E695-485A-B04C-3DD64B17318E

Device           Start     End Sectors Size Type
/dev/nvme0n2p1    2048 4196351 4194304   2G Linux filesystem
/dev/nvme0n2p2 4196352 8390655 4194304   2G Linux filesystem

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
```

```
vishal@localhost:~$ lsblk /dev/nvme0n2
NAME        MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n2     259:1    0  10G  0 disk
├─nvme0n2p1 259:5    0   2G  0 part
└─nvme0n2p2 259:6    0   2G  0 part
vishal@localhost:~$ sudo fdisk /dev/nvme0n2
[sudo] password for vishal:

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.


Command (m for help): d
Partition number (1,2, default 2): 2

Partition 2 has been deleted.
```

e. Configure systems to mount file systems at boot by universally unique ID (UUID) or label

```
vishal@localhost:~$ sudo blkid /dev/nvme0n2p1
[sudo] password for vishal:
/dev/nvme0n2p1: UUID="c669ffd7-dd89-42c6-9e43-b4d86d760913" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="d5f6549b-2099-4158-af49-cdde3bf95cec"
vishal@localhost:~$ sudo mkdir /mnt/ext4disk
mkdir: cannot create directory ‘/mnt/ext4disk’: File exists
vishal@localhost:~$ findmnt /mnt/ext4disk
vishal@localhost:~$ sudo nano /etc/fstab
vishal@localhost:~$ sudo mount -a
mount: (hint) your fstab has been modified, but systemd still uses
       the old version; use 'systemctl daemon-reload' to reload.
vishal@localhost:~$ df -h /mnt/ext4disk
Filesystem      Size  Used Avail Use% Mounted on
/dev/nvme0n2p1  2.0G   24K  1.8G   1% /mnt/ext4disk
```

f. Add new partitions and logical volumes, and swap to a system Non-destructively

```
vishal@localhost:~$ sudo fdisk /dev/nvme0n2

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.


Command (m for help): n
Partition number (2-128, default 2):
First sector (4196352-20971486, default 4196352):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (4196352-20971486, default 20969471):

Created a new partition 2 of type 'Linux filesystem' and of size 8 GiB.

Command (m for help): p
Disk /dev/nvme0n2: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual NVMe Disk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: C81BA5CF-E695-485A-B04C-3DD64B17318E

Device           Start      End  Sectors Size Type
/dev/nvme0n2p1    2048  4196351  4194304   2G Linux filesystem
/dev/nvme0n2p2 4196352 20969471 16773120   8G Linux filesystem
```

```
vishal@localhost:~$ sudo pvcreate /dev/nvme0n2p2
[sudo] password for vishal:
  Physical volume "/dev/nvme0n2p2" successfully created.
vishal@localhost:~$ sudo pvs
  PV             VG   Fmt  Attr PSize  PFree
  /dev/nvme0n1p3 rhel lvm2 a--  48.41g     0
  /dev/nvme0n2p2      lvm2 ---  <8.00g <8.00g
vishal@localhost:~$ sudo vgcreate practical7_vg /dev/nvme0n2p2
  Volume group "practical7_vg" successfully created
vishal@localhost:~$ sudo vgs
  VG            #PV #LV #SN Attr   VSize  VFree
  practical7_vg   1   0   0 wz--n- <8.00g <8.00g
  rhel            1   2   0 wz--n- 48.41g     0
vishal@localhost:~$ sudo lvcreate -L 2G -n practical7_lv practical7_vg
  Logical volume "practical7_lv" created.
vishal@localhost:~$ sudo lvs
  LV            VG            Attr       LSize  Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
  practical7_lv practical7_vg -wi-a-----  2.00g
  root          rhel          -wi-ao---- 44.50g
  swap          rhel          -wi-ao----  3.91g
vishal@localhost:~$ sudo lvcreate -L 2G -n practical7_lv practical7_vg
  Logical Volume "practical7_lv" already exists in volume group "practical7_vg"
vishal@localhost:~$ sudo lvs
  LV            VG            Attr       LSize  Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
  practical7_lv practical7_vg -wi-a-----  2.00g
  root          rhel          -wi-ao---- 44.50g
  swap          rhel          -wi-ao----  3.91g

vishal@localhost:~$ sudo lvcreate -L 1G -n practical7_swap practical7_vg
  Logical volume "practical7_swap" created.
vishal@localhost:~$ sudo mkswap /dev/practical7_vg/practical7_swap
Setting up swapspace version 1, size = 1024 MiB (1073737728 bytes)
no label, UUID=ed39cfa9-da39-4be6-b2dc-0f0cb7ad5ca0
vishal@localhost:~$ sudo swapon /dev/practical7_vg/practical7_swap
vishal@localhost:~$ swapon --show
NAME      TYPE       SIZE USED PRIO
/dev/dm-1 partition  3.9G   0B   -2
/dev/dm-3 partition 1024M   0B   -3
```

g. Create, mount, unmount, and use vfat, ext4, and xfs file systems

```
vishal@localhost:~$ which mkfs.vfat
which mkfs.ext4
which mkfs.xfs
/usr/sbin/mkfs.vfat
/usr/sbin/mkfs.ext4
/usr/sbin/mkfs.xfs
vishal@localhost:~$ sudo dd if=/dev/zero of=/vfat.img bs=1M count=500
[sudo] password for vishal:
500+0 records in
500+0 records out
524288000 bytes (524 MB, 500 MiB) copied, 1.23547 s, 424 MB/s
vishal@localhost:~$ sudo mkfs.vfat /vfat.img
mkfs.fat 4.2 (2021-01-31)
vishal@localhost:~$ sudo mkdir -p /mnt/vfat
vishal@localhost:~$ sudo mount -o loop /vfat.img /mnt/vfat
vishal@localhost:~$ df -Th /mnt/vfat
Filesystem     Type  Size  Used Avail Use% Mounted on
/dev/loop0     vfat  500M     0  500M   0% /mnt/vfat
vishal@localhost:~$ sudo touch /mnt/vfat/test.txt
vishal@localhost:~$ ls -l /mnt/vfat
total 0
-rwxr-xr-x. 1 root root 0 Sep 14 12:04 test.txt
vishal@localhost:~$ sudo umount /mnt/vfat
vishal@localhost:~$ lsblk
NAME                            MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sr0                              11:0    1  7.9G  0 rom  /run/media/vishal/RHEL-10-0-BaseOS-x86_64
nvme0n1                         259:0    0   50G  0 disk
├─nvme0n1p1                     259:2    0  600M  0 part /boot/efi
├─nvme0n1p2                     259:3    0    1G  0 part /boot
└─nvme0n1p3                     259:4    0 48.4G  0 part
  ├─rhel-root                   253:0    0 44.5G  0 lvm  /
  └─rhel-swap                   253:1    0  3.9G  0 lvm  [SWAP]
nvme0n2                         259:1    0   10G  0 disk
├─nvme0n2p1                     259:5    0    2G  0 part
└─nvme0n2p2                     259:6    0    8G  0 part
  ├─practical7_vg-practical7_lv 253:2    0    2G  0 lvm
  └─practical7_vg-practical7_swap 253:3    0    1G  0 lvm
```

```
vishal@localhost:~$ sudo dd if=/dev/zero of=/ext4.img bs=1M count=500
[sudo] password for vishal:
500+0 records in
500+0 records out
524288000 bytes (524 MB, 500 MiB) copied, 1.57242 s, 333 MB/s
vishal@localhost:~$ sudo mkfs.ext4 /ext4.img
mke2fs 1.47.1 (20-May-2024)
Discarding device blocks: done
Creating filesystem with 512000 1k blocks and 128016 inodes
Filesystem UUID: 34eee2a7-dc23-4795-8f3e-d06bbd567e3b
Superblock backups stored on blocks:
        8193, 24577, 40961, 57345, 73729, 204801, 221185, 401409

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

vishal@localhost:~$ sudo mkdir -p /mnt/ext4
vishal@localhost:~$ sudo mount -o loop /ext4.img /mnt/ext4
vishal@localhost:~$ df -Th /mnt/ext4
Filesystem     Type  Size  Used Avail Use% Mounted on
/dev/loop0     ext4  459M   14K  430M   1% /mnt/ext4
vishal@localhost:~$ sudo touch /mnt/ext4/test.txt
vishal@localhost:~$ sudo umount /mnt/ext4
vishal@localhost:~$ sudo dd if=/dev/zero of=/xfs.img bs=1M count=500
[sudo] password for vishal:
500+0 records in
500+0 records out
524288000 bytes (524 MB, 500 MiB) copied, 0.169123 s, 3.1 GB/s
vishal@localhost:~$ sudo mkfs.xfs /xfs.img
meta-data=/xfs.img               isize=512    agcount=4, agsize=32000 blks
         =                       sectsz=512   attr=2, projid32bit=1
         =                       crc=1        finobt=1, sparse=1, rmapbt=1
         =                       reflink=1    bigtime=1 inobtcount=1 nrext64=1
         =                       exchange=0
data     =                       bsize=4096   blocks=128000, imaxpct=25
         =                       sunit=0      swidth=0 blks
naming   =version 2              bsize=4096   ascii-ci=0, ftype=1, parent=0
log      =internal log           bsize=4096   blocks=16384, version=2
         =                       sectsz=512   sunit=0 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
vishal@localhost:~$ sudo mkdir /mnt/xfs
vishal@localhost:~$ sudo mount -o loop /xfs.img /mnt/xfs
vishal@localhost:~$ df -Th /mnt/xfs
Filesystem     Type  Size  Used Avail Use% Mounted on
/dev/loop0     xfs   436M   34M  403M   8% /mnt/xfs

vishal@localhost:~$ sudo touch /mnt/xfs/test.txt
vishal@localhost:~$ sudo umount /mnt/xfs
```

Practical 8

Aim: Configuring Networking

a. Managing Network Addresses and Interfaces

```
vishal@localhost:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:44:d9:0b brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    altname enx000c2944d90b
    inet 192.168.105.128/24 brd 192.168.105.255 scope global dynamic noprefixroute ens160
       valid_lft 1672sec preferred_lft 1672sec
    inet6 fe80::20c:29ff:fe44:d90b/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
vishal@localhost:~$ nmcli device status
DEVICE  TYPE      STATE                   CONNECTION
ens160  ethernet  connected               ens160
lo      loopback  connected (externally)  lo
vishal@localhost:~$ nmcli connection show
NAME    UUID                                  TYPE      DEVICE
ens160  1031a884-2779-3b25-9a84-f104d2636e9d  ethernet  ens160
lo      5ee32bb0-6e86-4d76-9754-81ef5235a664  loopback  lo
```

b. Validating Network Configuration

```
vishal@localhost:~$ ip route
default via 192.168.105.2 dev ens160 proto dhcp src 192.168.105.128 metric 100
192.168.105.0/24 dev ens160 proto kernel scope link src 192.168.105.128 metric 100
vishal@localhost:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=50.6 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=14.0 ms

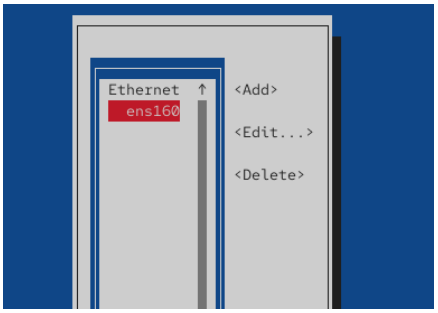
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 2 received, 50% packet loss, time 3053ms
rtt min/avg/max/mdev = 14.016/32.292/50.569/18.276 ms
vishal@localhost:~$ ping -c 4 google.com
PING google.com (142.250.134.138) 56(84) bytes of data.
64 bytes from fx-in-f138.1e100.net (142.250.134.138): icmp_seq=1 ttl=128 time=11.4 ms
64 bytes from fx-in-f138.1e100.net (142.250.134.138): icmp_seq=2 ttl=128 time=5.92 ms
64 bytes from fx-in-f138.1e100.net (142.250.134.138): icmp_seq=3 ttl=128 time=13.8 ms
64 bytes from fx-in-f138.1e100.net (142.250.134.138): icmp_seq=4 ttl=128 time=5.95 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 5.920/9.270/13.844/3.450 ms
```

c. Managing Network Configuration with nmtui and nmcli

```
vishal@localhost:~$ sudo nmtui
[sudo] password for vishal:
```





```
vishal@localhost:~$ nmcli connection show ens160
connection.id:                          ens160
connection.uuid:                        1031a884-2779-3b25-9a84-f104d2636e9d
connection.stable-id:                   --
connection.type:                        802-3-ethernet
connection.interface-name:              ens160
connection.autoconnect:                 yes
connection.autoconnect-priority:        -999
connection.autoconnect-retries:         -1 (default)
connection.multi-connect:               0 (default)
connection.auth-retries:                -1
connection.timestamp:                   1789367109
connection.permissions:                 --
connection.zone:                        --
connection.controller:                  --
connection.master:                      --
connection.slave-type:                  --
connection.port-type:                   --
connection.autoconnect-slaves:          -1 (default)
connection.autoconnect-ports:           -1 (default)
connection.down-on-poweroff:            -1 (default)
connection.secondaries:                 --
connection.gateway-ping-timeout:        0
connection.ip-ping-timeout:             0
connection.ip-ping-addresses:           --
connection.ip-ping-addresses-require-all:-1 (default)
connection.metered:                     unknown
connection.lldp:                        default
connection.mdns:                        -1 (default)
connection.llmnr:                       -1 (default)
connection.dns-over-tls:                -1 (default)
connection.mptcp-flags:                 0x0 (default)
connection.wait-device-timeout:         -1
connection.wait-activation-delay:       -1
802-3-ethernet.port:                    --
802-3-ethernet.speed:                   0
802-3-ethernet.duplex:                  --
802-3-ethernet.auto-negotiate:          no
802-3-ethernet.mac-address:             --
802-3-ethernet.cloned-mac-address:      --
802-3-ethernet.generate-mac-address-mask:--
802-3-ethernet.mac-address-denylist:    --

ipv4.dns-search:                            --
ipv4.dns-options:                           --
ipv4.dns-priority:                          0
ipv4.addresses:                             --
ipv4.gateway:                               --
ipv4.routes:                                --
ipv4.route-metric:                          -1
ipv4.route-table:                           0 (unspec)
ipv4.routing-rules:                         --
ipv4.replace-local-rule:                    -1 (default)
ipv4.dhcp-send-release:                     -1 (default)
ipv4.routed-dns:                            -1 (default)
ipv4.ignore-auto-routes:                    no
ipv4.ignore-auto-dns:                       no
ipv4.dhcp-client-id:                        --
ipv4.dhcp-iaid:                             --
ipv4.dhcp-dscp:                             --
ipv4.dhcp-timeout:                          0 (default)
ipv4.dhcp-send-hostname-deprecated:         yes
ipv4.dhcp-send-hostname:                    -1 (default)
ipv4.dhcp-hostname:                         --
ipv4.dhcp-fqdn:                             --
ipv4.dhcp-hostname-flags:                   0x0 (none)
ipv4.never-default:                         no
ipv4.may-fail:                              yes
ipv4.required-timeout:                      -1 (default)
ipv4.dad-timeout:                           -1 (default)
ipv4.dhcp-vendor-class-identifier:          --
ipv4.dhcp-ipv6-only-preferred:              -1 (default)
ipv4.link-local:                            0 (default)
ipv4.dhcp-reject-servers:                   --
ipv4.auto-route-ext-gw:                     -1 (default)
ipv4.shared-dhcp-range:                     --
ipv4.shared-dhcp-lease-time:                0 (default)
ipv6.method:                                auto
ipv6.dns:                                   --
```

d. Configure IPv4 and IPv6 addresses

```
vishal@localhost:~$ sudo nmcli connection modify ens160 ipv4.method manual ipv4.addresses 192.168.105.150/24 ipv4.gateway 192.168.105.2
[sudo] password for vishal:
vishal@localhost:~$ sudo nmcli connection modify ens160 ipv4.dns 8.8.8.8
vishal@localhost:~$ sudo nmcli connection down ens160
sudo nmcli connection up ens160
Connection 'ens160' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/4)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/5)
vishal@localhost:~$ ip addr show ens160
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:44:d9:0b brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    altname enx000c2944d90b
    inet 192.168.105.150/24 brd 192.168.105.255 scope global noprefixroute ens160
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe44:d90b/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
vishal@localhost:~$ sudo nmcli connection modify ens160 ipv6.method manual ipv6.addresses 2001:db8::10/64
vishal@localhost:~$ sudo nmcli connection down ens160
sudo nmcli connection up ens160
Connection 'ens160' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/5)
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/6)
vishal@localhost:~$ ip -6 addr show ens160
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    altname enp3s0
    altname enx000c2944d90b
    inet6 2001:db8::10/64 scope global noprefixroute
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe44:d90b/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

e. Configure DNS-hostname resolution

```
vishal@localhost:~$ hostname
localhost.localdomain
vishal@localhost:~$ sudo hostnamectl set-hostname rhel.local
vishal@localhost:~$ hostname
rhel.local
vishal@localhost:~$ getent hosts google.com
2404:6800:4000:1017::65 google.com
2404:6800:4000:1017::8b google.com
2404:6800:4000:1017::64 google.com
2404:6800:4000:1017::71 google.com
vishal@localhost:~$ hostnamectl
 Static hostname: rhel.local
       Icon name: computer-vm
         Chassis: vm 🖴
      Machine ID: 088301b180f240dbb26272fd219c3676
         Boot ID: a23fff6e87fa43c59a8b6de3b73cc89a
    AF_VSOCK CID: 4265924875
  Virtualization: vmware
Operating System: Red Hat Enterprise Linux 10.0 (Coughlan)
     CPE OS Name: cpe:/o:redhat:enterprise_linux:10::baseos
          Kernel: Linux 6.12.0-55.9.1.el10_0.x86_64
    Architecture: x86-64
 Hardware Vendor: VMware, Inc.
  Hardware Model: VMware20,1
Firmware Version: VMW201.00V.25275966.B64.2603102050
   Firmware Date: Tue 2026-03-10
    Firmware Age: 6month 5d
vishal@localhost:~$ getent hosts rhel.local
2001:db8::10    rhel.local
fe80::20c:29ff:fe44:d90b rhel.local
```

Practical 9

Aim: Configuring SSH

a. Using SSH and Related Utilities

```
vishal@localhost:~$ ssh -V
OpenSSH_9.9p1, OpenSSL 3.2.2 4 Jun 2024
vishal@localhost:~$ sudo systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-08-25 21:41:32 IST; 2 weeks 5 days ago
 Invocation: c856ae7791ad4e528641aa8a6fadd894
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 1128 (sshd)
      Tasks: 1 (limit: 22706)
     Memory: 1.8M (peak: 2.1M)
        CPU: 44ms
     CGroup: /system.slice/sshd.service
             └─1128 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 25 21:41:32 localhost.localdomain systemd[1]: Starting sshd.service - OpenSSH server daemon...
Aug 25 21:41:32 localhost.localdomain (sshd)[1128]: sshd.service: Referenced but unset environment variable evaluates to an empty string: OPTIONS
Aug 25 21:41:32 localhost.localdomain sshd[1128]: Server listening on 0.0.0.0 port 22.
Aug 25 21:41:32 localhost.localdomain sshd[1128]: Server listening on :: port 22.
Aug 25 21:41:32 localhost.localdomain systemd[1]: Started sshd.service - OpenSSH server daemon.
vishal@localhost:~$ sudo systemctl start sshd
vishal@localhost:~$ sudo systemctl enable sshd
vishal@localhost:~$ which ssh
which ssh-keygen
which scp
/usr/bin/ssh
/usr/bin/ssh-keygen
/usr/bin/scp
```

b. Access remote systems using SSH

```
vishal@localhost:~$ ssh localhost
The authenticity of host 'localhost (::1)' can't be established.
ED25519 key fingerprint is SHA256:t/P0Rl9V5dV4aPFV2O/esEPN/iUYGHLYp1S+WGgMA9I.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
vishal@localhost's password:
Web console: https://localhost:9090/ or https://192.168.105.128:9090/

Register this system with Red Hat Insights: rhc connect

Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Insights will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://red.ht/registration
Last login: Tue Aug 25 21:42:40 2026
vishal@rhel:~$ whoami
vishal
vishal@rhel:~$ hostname
rhel.local
```

c. Configuring Key-Based Authentication with Passphrases

```
vishal@localhost:~$ ssh-keygen -t ed25519
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/vishal/.ssh/id_ed25519):
Enter passphrase for "/home/vishal/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/vishal/.ssh/id_ed25519
Your public key has been saved in /home/vishal/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:Kn4r2cQ+SVgunYtoh657CwE56DIqUIRhaYaH7ko0hTI vishal@rhel.local
The key's randomart image is:
+--[ED25519 256]--+
|o=+              |
|EB..             |
|Xoo              |
|o*    .          |
|B..  * .S        |
|+=  o B.         |
|*  o.O.o         |
|o.=.=.B          |
|o*oo.o.o         |
+----[SHA256]-----+
vishal@localhost:~$ ssh-copy-id vishal@localhost
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ssh-add -L
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
vishal@localhost's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'vishal@localhost'"
and check to make sure that only the key(s) you wanted were added.

vishal@localhost:~$ ssh vishal@localhost
Web console: https://localhost:9090/ or https://192.168.105.128:9090/

Register this system with Red Hat Insights: rhc connect

Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Insights will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://red.ht/registration
Last login: Mon Sep 14 12:58:00 2026 from ::1
vishal@rhel:~$ exit
```

d. Configure key-based authentication for SSH

```
vishal@localhost:~$ ls -l ~/.ssh/
total 20
-rw-------. 1 vishal vishal  99 Sep 14 13:01 authorized_keys
-rw-------. 1 vishal vishal 411 Sep 14 13:01 id_ed25519
-rw-r--r--. 1 vishal vishal  99 Sep 14 13:01 id_ed25519.pub
-rw-------. 1 vishal vishal 825 Sep 14 12:58 known_hosts
-rw-r--r--. 1 vishal vishal  91 Sep 14 12:57 known_hosts.old
vishal@localhost:~$ cat ~/.ssh/authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIFb9W2I0zMWl6AYWzLvWTTIcGCJ/XkmStjazxSuuE+1F vishal@rhel.local
vishal@localhost:~$ chmod 700 ~/.ssh
chmod 600 ~/.ssh/authorized_keys
vishal@localhost:~$ ssh vishal@localhost
Web console: https://localhost:9090/ or https://192.168.105.128:9090/

Register this system with Red Hat Insights: rhc connect

Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Insights will enable analytics and additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/insights

You can learn more about how to register your system
using rhc at https://red.ht/registration
Last login: Mon Sep 14 13:02:08 2026 from ::1
```

Practical 10

Aim: Configuring web server

a. Configuring a Basic Apache Server

```
vishal@rhel:~$ sudo systemctl enable --now httpd
Created symlink '/etc/systemd/system/multi-user.target.wants/httpd.service' → '/usr/lib/systemd/system/httpd.service'.
vishal@rhel:~$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-09-14 13:47:53 IST; 13s ago
 Invocation: 77da8298c77f437bbf784f0708382c0d
       Docs: man:httpd.service(8)
   Main PID: 9067 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 22706)
     Memory: 14.1M (peak: 14.4M)
        CPU: 761ms
     CGroup: /system.slice/httpd.service
             ├─9067 /usr/sbin/httpd -DFOREGROUND
             ├─9068 /usr/sbin/httpd -DFOREGROUND
             ├─9069 /usr/sbin/httpd -DFOREGROUND
             ├─9070 /usr/sbin/httpd -DFOREGROUND
             └─9071 /usr/sbin/httpd -DFOREGROUND

Sep 14 13:47:52 rhel.local systemd[1]: Starting httpd.service - The Apache HTTP Server...
Sep 14 13:47:53 rhel.local (httpd)[9067]: httpd.service: Referenced but unset environment variable evaluates to an empty string: OPTIONS
Sep 14 13:47:53 rhel.local systemd[1]: Started httpd.service - The Apache HTTP Server.
Sep 14 13:47:53 rhel.local httpd[9067]: Server configured, listening on: port 80
vishal@rhel:~$ sudo nano /var/www/html/index.html
vishal@rhel:~$ curl http://localhost
<h1>Hello from RHEL Apache Server</h1>
<p>Apache Web Server is working.</p>
```

b. Understanding Apache Configuration Files

```
vishal@rhel:~$ ls /etc/httpd/
conf  conf.d  conf.modules.d  logs  modules  run  state
vishal@rhel:~$ ls /etc/httpd/conf.d/
autoindex.conf  README  userdir.conf  welcome.conf
vishal@rhel:~$ cat /etc/httpd/conf/httpd.conf
#
# This is the main Apache HTTP server configuration file.  It contains the
# configuration directives that give the server its instructions.
# See <URL:http://httpd.apache.org/docs/2.4/> for detailed information.
# In particular, see
# <URL:http://httpd.apache.org/docs/2.4/mod/directives.html>
# for a discussion of each configuration directive.
#
# See the httpd.conf(5) man page for more information on this configuration,
# and httpd.service(8) on using and configuring the httpd service.
#
# Do NOT simply read the instructions in here without understanding
# what they do.  They're here only as hints or reminders.  If you are unsure
# consult the online docs. You have been warned.
#
# Configuration and logfile names: If the filenames you specify for many
# of the server's control files begin with "/" (or "drive:/" for Win32), the
# server will use that explicit path.  If the filenames do *not* begin
# with "/", the value of ServerRoot is prepended -- so 'log/access_log'
# with ServerRoot set to '/www' will be interpreted by the
# server as '/www/log/access_log', where as '/log/access_log' will be
# interpreted as '/log/access_log'.

#
# ServerRoot: The top of the directory tree under which the server's
# configuration, error, and log files are kept.
#
# Do not add a slash at the end of the directory path.  If you point
# ServerRoot at a non-local disk, be sure to specify a local disk on the
# Mutex directive, if file-based mutexes are used.  If you wish to share the
# same ServerRoot for multiple httpd daemons, you will need to change at
# least PidFile.
#
ServerRoot "/etc/httpd"

#
# Listen: Allows you to bind Apache to specific IP addresses and/or
# ports, instead of the default. See also the <VirtualHost>
# directive.
#
# Change this to Listen on a specific IP address, but note that if
# httpd.service is enabled to run at boot time, the address may not be
# available when the service starts.  See the httpd.service(8) man
# page for more information.
#
#Listen 12.34.56.78:80
Listen 80
```

c. Creating Apache Virtual Host

```
vishal@rhel:~$ sudo nano /etc/httpd/conf.d/vhost.conf
[sudo] password for vishal:
vishal@rhel:~$ sudo mkdir -p /var/www/mysite
[sudo] password for vishal:
vishal@rhel:~$ sudo nano /var/www/mysite/index.html
vishal@rhel:~$ sudo apachectl configtest
Syntax OK
vishal@rhel:~$ sudo systemctl restart httpd
vishal@rhel:~$ sudo nano /etc/hosts
vishal@rhel:~$ curl http://mysite.local
<h1>Welcome to My Virtual Host</h1>
```

Practical 11

Aim: Configure Firewall

a. Linux Firewall with iptables

```
vishal@rhel:~$ sudo iptables --version
[sudo] password for vishal:
iptables v1.8.11 (nf_tables)
vishal@rhel:~$ sudo iptables -L -n -v
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
vishal@rhel:~$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
vishal@rhel:~$ sudo iptables -L -n -v
Chain INPUT (policy ACCEPT 1 packets, 76 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 ACCEPT     tcp  --  *      *       0.0.0.0/0            0.0.0.0/0           tcp dpt:22

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
```

b. Working with Firewalld

```
vishal@rhel:~$ sudo systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
     Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-08-25 21:41:31 IST; 2 weeks 5 days ago
 Invocation: d1ab5369a9d14aad9344800700cfa010
       Docs: man:firewalld(1)
   Main PID: 1038 (firewalld)
      Tasks: 2 (limit: 22706)
     Memory: 47.7M (peak: 49.5M)
        CPU: 1.699s
     CGroup: /system.slice/firewalld.service
             └─1038 /usr/bin/python3 -sP /usr/sbin/firewalld --nofork --nopid

Aug 25 21:41:29 localhost systemd[1]: Starting firewalld.service - firewalld - dynamic firewall daemon...
Aug 25 21:41:31 localhost systemd[1]: Started firewalld.service - firewalld - dynamic firewall daemon.
vishal@rhel:~$ sudo systemctl enable --now firewalld
vishal@rhel:~$ sudo firewall-cmd --get-active-zones
public (default)
  interfaces: ens160
vishal@rhel:~$ sudo firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: ens160
  sources:
  services: cockpit dhcpv6-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

Practical 12

Aim: Accessing Network Storage

a. Using NFS Services

```
vishal@rhel:~$ sudo mkdir -p /srv/nfs/share
sudo chmod 777 /srv/nfs/share
vishal@rhel:~$ sudo nano /etc/exports
vishal@rhel:~$ sudo systemctl enable --now nfs-server
Created symlink '/etc/systemd/system/multi-user.target.wants/nfs-server.service' → '/usr/lib/systemd/system/nfs-server.service'.
vishal@rhel:~$ sudo exportfs -rav
exporting 127.0.0.1:/srv/nfs/share
vishal@rhel:~$ sudo exportfs -v
/srv/nfs/share  127.0.0.1(sync,wdelay,hide,no_subtree_check,sec=sys,rw,secure,root_squash,no_all_squash)
vishal@rhel:~$ sudo mkdir -p /mnt/nfs
vishal@rhel:~$ sudo mount -t nfs 127.0.0.1:/srv/nfs/share /mnt/nfs
vishal@rhel:~$ df -Th /mnt/nfs
Filesystem               Type  Size  Used Avail Use% Mounted on
127.0.0.1:/srv/nfs/share nfs4   45G  5.7G   39G  13% /mnt/nfs
vishal@rhel:~$ sudo touch /mnt/nfs/test.txt
ls -l /mnt/nfs
total 0
-rw-r--r--. 1 nobody nobody 0 Sep 14 14:36 test.txt
vishal@rhel:~$ sudo umount /mnt/nfs
```

b. Using CIFS Services

```
vishal@rhel:~$ sudo mkdir -p /mnt/cifs
vishal@rhel:~$ sudo dnf install -y samba samba-client
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "subscription-manager" to register.

Last metadata expiration check: 2:12:26 ago on Monday 14 September 2026 12:27:01 PM.
Dependencies resolved.
================================================================================================================================================================================
 Package                                  Architecture                    Version                                  Repository                                          Size
================================================================================================================================================================================
Installing:
 samba                                    x86_64                          4.21.3-102.el10                          BaseOS                                             986 k
 samba-client                             x86_64                          4.21.3-102.el10                          AppStream                                          752 k
Installing dependencies:
 libnetapi                                x86_64                          4.21.3-102.el10                          BaseOS                                             148 k
 samba-common-tools                       x86_64                          4.21.3-102.el10                          BaseOS                                             486 k
 samba-dcerpc                             x86_64                          4.21.3-102.el10                          BaseOS                                             724 k
 samba-ldb-ldap-modules                   x86_64                          4.21.3-102.el10                          BaseOS                                              39 k
 samba-libs                               x86_64                          4.21.3-102.el10                          BaseOS                                             132 k

Transaction Summary
================================================================================================================================================================================
Install  7 Packages

Total size: 3.2 M
Installed size: 11 M
Downloading Packages:
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                                                                       1/1
  Installing       : samba-libs-4.21.3-102.el10.x86_64                                                                                                                     1/7
  Installing       : libnetapi-4.21.3-102.el10.x86_64                                                                                                                      2/7
  Installing       : samba-dcerpc-4.21.3-102.el10.x86_64                                                                                                                   3/7
  Installing       : samba-ldb-ldap-modules-4.21.3-102.el10.x86_64                                                                                                         4/7
  Installing       : samba-common-tools-4.21.3-102.el10.x86_64                                                                                                             5/7
  Installing       : samba-4.21.3-102.el10.x86_64                                                                                                                          6/7
  Running scriptlet: samba-4.21.3-102.el10.x86_64                                                                                                                          6/7
  Installing       : samba-client-4.21.3-102.el10.x86_64                                                                                                                   7/7
  Running scriptlet: samba-client-4.21.3-102.el10.x86_64                                                                                                                   7/7
Installed products updated.

Installed:
  libnetapi-4.21.3-102.el10.x86_64   samba-4.21.3-102.el10.x86_64   samba-client-4.21.3-102.el10.x86_64   samba-common-tools-4.21.3-102.el10.x86_64   samba-dcerpc-4.21.3-102.el10.x86_64   samba-ldb-ldap-modules-4.21.3-102.el10.x86_64
  samba-libs-4.21.3-102.el10.x86_64

Complete!
vishal@rhel:~$ sudo mkdir -p /srv/cifs/share
sudo chmod 777 /srv/cifs/share
vishal@rhel:~$ sudo nano /etc/samba/smb.conf
vishal@rhel:~$ sudo systemctl enable --now smb
Created symlink '/etc/systemd/system/multi-user.target.wants/smb.service' → '/usr/lib/systemd/system/smb.service'.
```

```
vishal@rhel:~$ sudo testparm
Load smb config files from /etc/samba/smb.conf
Loaded services file OK.
Weak crypto is allowed by GnuTLS (e.g. NTLM as a compatibility fallback)

Server role: ROLE_STANDALONE

Press enter to see a dump of your service definitions

# Global parameters
[global]
        printcap name = cups
        security = USER
        workgroup = SAMBA
        idmap config * : backend = tdb
        cups options = raw
        include = /etc/samba/usershares.conf


[homes]
        browseable = No
        comment = Home Directories
        inherit acls = Yes
        read only = No
        valid users = %S %D%w%S


[printers]
        browseable = No
        comment = All Printers
        create mask = 0600
        path = /var/tmp
        printable = Yes


[print$]
        comment = Printer Drivers
        create mask = 0664
        directory mask = 0775
        force group = @printadmin
        path = /var/lib/samba/drivers
        write list = @printadmin root


[cifsshare]
        guest ok = Yes
        path = /srv/cifs/share
        read only = No
vishal@rhel:~$ sudo mount -t cifs //127.0.0.1/cifsshare /mnt/cifs -o guest
vishal@rhel:~$ df -Th /mnt/cifs
Filesystem            Type  Size  Used Avail Use% Mounted on
//127.0.0.1/cifsshare cifs   45G  5.7G   39G  13% /mnt/cifs
vishal@rhel:~$ sudo touch /mnt/cifs/test.txt
ls -l /mnt/cifs
touch: cannot touch '/mnt/cifs/test.txt': Permission denied
total 0
vishal@rhel:~$ sudo umount /mnt/cifs
```

c. Mounting Remote File Systems Through fstab

```
vishal@rhel:~$ sudo mount -t cifs //127.0.0.1/cifsshare /mnt/cifs -o guest
vishal@rhel:~$ df -Th /mnt/cifs
Filesystem            Type  Size  Used Avail Use% Mounted on
//127.0.0.1/cifsshare cifs   45G  5.7G   39G  13% /mnt/cifs
vishal@rhel:~$ sudo touch /mnt/cifs/test.txt
ls -l /mnt/cifs
touch: cannot touch '/mnt/cifs/test.txt': Permission denied
total 0
vishal@rhel:~$ sudo umount /mnt/cifs
vishal@rhel:~$ sudo mkdir -p /mnt/nfs
vishal@rhel:~$ sudo nano /etc/fstab
vishal@rhel:~$ sudo mount -a
mount: (hint) your fstab has been modified, but systemd still uses
       the old version; use 'systemctl daemon-reload' to reload.
vishal@rhel:~$ df -Th /mnt/nfs
Filesystem              Type  Size  Used Avail Use% Mounted on
127.0.0.1:/srv/nfs/share nfs4   45G  5.7G   39G  13% /mnt/nfs
```

d. Using Automount to Mount Remote File Systems

```
vishal@rhel:~$ sudo dnf install -y autofs
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "subscription-manager" to register.

Last metadata expiration check: 2:24:42 ago on Monday 14 September 2026 12:27:01 PM.
Dependencies resolved.
================================================================================
 Package                 Architecture          Version                Repository      Size
================================================================================
Installing:
 autofs                  x86_64                1:5.1.9-11.el10        BaseOS         395 k
Installing dependencies:
 libsss_autofs           x86_64                2.10.2-3.el10          BaseOS          37 k

Transaction Summary
================================================================================
Install  2 Packages

Total size: 432 k
Installed size: 1.1 M
Downloading Packages:
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Installing       : libsss_autofs-2.10.2-3.el10.x86_64                     1/2
  Installing       : autofs-1:5.1.9-11.el10.x86_64                          2/2
  Running scriptlet: autofs-1:5.1.9-11.el10.x86_64                          2/2
Installed products updated.

Installed:
  autofs-1:5.1.9-11.el10.x86_64                libsss_autofs-2.10.2-3.el10.x86_64

Complete!
vishal@rhel:~$ sudo mkdir -p /mnt/auto
vishal@rhel:~$ sudo nano /etc/auto.master
vishal@rhel:~$ sudo nano /etc/auto.nfs
vishal@rhel:~$ sudo systemctl enable --now autofs
Created symlink '/etc/systemd/system/multi-user.target.wants/autofs.service' → '/usr/lib/systemd/system/autofs.service'.
vishal@rhel:~$ ls /mnt/auto/share
test.txt
```

Practical 13

Aim: Creating Shell scripts

a. automate or monitor tasks/processes

```
vishal@rhel:~$ nano task_monitor.sh
vishal@rhel:~$ chmod +x task_monitor.sh
vishal@rhel:~$ ./task_monitor.sh
Current Date and Time:
Monday 14 September 2026 02:22:02 PM IST
Logged-in User:
vishal
System Uptime:
 14:22:02 up  4:18,  2 users,  load average: 0.71, 0.61, 0.62
Running Processes:
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.3  1.1  49188 41316 ?        Ss   10:04   0:52 /usr/lib/systemd/systemd --switched-root --system --deserialize=48 rhgb
root           2  0.0  0.0      0     0 ?        S    10:04   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    10:04   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   10:04   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   10:04   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   10:04   0:00 [kworker/R-slub_flushwq]
root           7  0.0  0.0      0     0 ?        I<   10:04   0:00 [kworker/R-netns]
root          10  0.0  0.0      0     0 ?        I<   10:04   0:00 [kworker/0:0H-events_highpri]
root          11  0.0  0.0      0     0 ?        I    10:04   0:00 [kworker/u512:0-events_unbound]
```

b. Getting and setting system information

```
vishal@rhel:~$ nano system_info.sh
vishal@rhel:~$ chmod +x system_info.sh
vishal@rhel:~$ ./system_info.sh
Hostname:
rhel.local
Kernel Version:
6.12.0-55.9.1.el10_0.x86_64
Operating System:
PRETTY_NAME="Red Hat Enterprise Linux 10.0 (Coughlan)"
Current User:
vishal
System Architecture:
x86_64
```

Practical 14

Aim: Scheduling Tasks

a. Configuring Cron to Automate Recurring Tasks

```
vishal@rhel:~$ systemctl status crond
● crond.service - Command Scheduler
     Loaded: loaded (/usr/lib/systemd/system/crond.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-08-25 21:41:32 IST; 2 weeks 5 days ago
 Invocation: b15a047769d44f458c37467d8bd1723b
   Main PID: 1136 (crond)
      Tasks: 1 (limit: 22706)
     Memory: 1.2M (peak: 3.4M)
        CPU: 678ms
     CGroup: /system.slice/crond.service
             └─1136 /usr/sbin/crond -n

Sep 14 12:01:01 localhost.localdomain anacron[5893]: Jobs will be executed sequentially
Sep 14 12:01:01 localhost.localdomain run-parts[5895]: (/etc/cron.hourly) finished 0anacron
Sep 14 12:37:01 localhost.localdomain anacron[5893]: Job `cron.daily' started
Sep 14 12:37:01 localhost.localdomain anacron[5893]: Job `cron.daily' terminated
Sep 14 12:57:01 rhel.local anacron[5893]: Job `cron.weekly' started
Sep 14 12:57:01 rhel.local anacron[5893]: Job `cron.weekly' terminated
Sep 14 12:57:01 rhel.local anacron[5893]: Normal exit (2 jobs run)
Sep 14 13:01:01 rhel.local CROND[7601]: (root) CMD (run-parts /etc/cron.hourly)
Sep 14 13:01:01 rhel.local run-parts[7604]: (/etc/cron.hourly) starting 0anacron
Sep 14 13:01:01 rhel.local run-parts[7610]: (/etc/cron.hourly) finished 0anacron
vishal@rhel:~$ sudo systemctl start crond
[sudo] password for vishal:
vishal@rhel:~$ sudo systemctl enable crond
vishal@rhel:~$ crontab -e
no crontab for vishal - using an empty one
crontab: installing new crontab
vishal@rhel:~$ crontab -l
* * * * * echo "Cron test" >> /home/vishal/cron_test.txt
vishal@rhel:~$ cat /home/vishal/cron_test.txt
Cron test
Cron test
Cron test
```

b. Configuring At to Schedule Future Tasks

```
vishal@rhel:~$ at -V
at version 3.2.5
vishal@rhel:~$ sudo systemctl enable --now atd
[sudo] password for vishal:
vishal@rhel:~$ echo "echo 'AT test' >> /home/vishal/at_test.txt" | at now + 2 minutes
warning: commands will be executed using /bin/sh
job 1 at Mon Sep 14 13:38:00 2026
vishal@rhel:~$ atq
1       Mon Sep 14 13:38:00 2026 a vishal
vishal@rhel:~$ cat /home/vishal/at_test.txt
AT test
```

c. Schedule tasks using at and cron

```
vishal@rhel:~$ crontab -l
* * * * * echo "Cron test" >> /home/vishal/cron_test.txt
vishal@rhel:~$ atq
vishal@rhel:~$ cat /home/vishal/cron_test.txt
Cron test
Cron test
Cron test
Cron test
Cron test
Cron test
Cron test
Cron test
Cron test
Cron test
Cron test
Cron test
Cron test
Cron test
Cron test
vishal@rhel:~$ cat /home/vishal/at_test.txt
AT test
```